



DEPARTMENT OF ICT

DIPLOMA IN INFORMATION TECHNOLOGY

YEAR 2 INTEGRATED PROJECT

PROJECT PROPOSAL

Project Title: Apex — Patient Queue & Records Management System

Group Name: Apex Systems

Student Number	Student Name	Role
22202248	Mtungwa ST	Project Manager
22418646	Mdubo N	System Analyst
22564960	Jongihlati SA	Developer
22412713	Mathebula A	Tester
22339104	Zulu Y	Documentation Lead
22518728	Khwela JM	UI/UX Designer

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1. Executive Summary

Apex is a proposed Windows-based desktop information system designed to address the persistent crisis in patient management at public healthcare facilities across South Africa. Clinics and hospitals — particularly those serving high-density communities — are overwhelmed by manual, paper-based processes that are inefficient, error-prone, and deeply frustrating for both patients and healthcare staff.

The proposed system will provide a centralised platform for registering patients, managing queues intelligently, and maintaining accessible digital patient records. Upon arrival, patients are registered and automatically assigned a queue number. Staff can monitor the queue dashboard, call the next patient, flag emergency or priority cases, and track waiting times. Patient records are stored digitally and are instantly searchable, meaning no file is ever lost and every visit is logged to a persistent history.

Apex does not require internet connectivity, making it viable for facilities in areas with unreliable network infrastructure. The system aims to reduce patient waiting times, eliminate lost records, improve the quality of care for patients with chronic conditions, and reduce administrative workload for healthcare staff.

2. Problem Statement

Public healthcare facilities across South Africa face a growing crisis in patient management due to manual, paper-based processes.

Who experiences the problem?

- Patients
- Receptionist and front-desk staff
- Nurses and clinicians
- Clinic and hospital administrators

What difficulties are caused?

- Disorganised waiting areas with no structured queue system
- Unpredictable and excessive patient waiting times (three to five hours in many cases)
- Lost, misplaced, or incomplete patient files
- Inability to retrieve patient history at the point of care
- Re-capturing of information from scratch due to missing records
- Increased administrative workload on already stretched staff

Why should the problem be solved?

The problem should be solved to reduce patient waiting times, prevent the loss of critical medical records, improve the quality of care for patients with chronic conditions, and ensure that healthcare facilities operate efficiently and effectively.

3. Current Situation

Many public healthcare facilities currently use paper files, physical folders, handwritten registers, and manual queue management to process patients.

Existing Methods

- Paper-based patient registration forms
- Physical file folders for patient records
- Handwritten appointment and queue registers
- Manual queue calling by reception staff

Current Challenges

- Files can be lost, misplaced, or damaged
- No visibility into queue order or estimated waiting time for patients
- Healthcare workers cannot quickly retrieve patient history
- Re-capturing of patient information wastes staff time

Limitations

- Slow access to patient information
- No reporting capability on patient volumes or waiting times
- Inconsistent record keeping across visits
- Poor tracking of chronic condition history and medication records

4. Proposed Solution

The proposed solution is Apex — a Windows-based desktop information system that digitises and streamlines patient flow and record management at healthcare facilities.

Main Purpose

To provide an efficient and centralised platform for registering patients, managing queues intelligently, and maintaining accessible digital patient records — eliminating the chaos of paper-based systems.

Key Features

- Patient registration and profile management
- Automated queue number assignment on arrival
- Dashboard with waiting time tracking
- Priority and emergency flagging
- Digital patient record storage and retrieval by name or ID number
- Persistent visit history per patient
- Report generation for patient volumes and operational statistics
- Offline operation — no internet connectivity required

Expected Benefits

- Reduced patient waiting times
- Elimination of lost or misplaced patient files
- Faster retrieval of patient records at the point of care
- Improved quality of care for patients with chronic conditions
- Reduced administrative workload for reception and nursing staff

5. Stakeholders

Stakeholder	Role in System
Clinic/Hospital Administrator	Oversees operations and reviews reports
Receptionist / Front-Desk Staff	Registers patients and manages the queue
Nurse / Clinician	Retrieves patient records and updates visit history
Patient	Receives a queue number and is called for service
Project Team	Develops and maintains the system

6. Project Objectives

1. Reduce patient waiting times through structured and visible queue management.
2. Eliminate lost or inaccessible patient records by digitalising file storage.
3. Improve retrieval time of patient history and records at the point of care.
4. Maintain accurate and persistent visit histories for each patient.
5. Generate administrative reports on patient volumes and facility operations.

7. Preliminary Scope

In Scope

6. Patient registration and profile management.
7. Queue number assignment and queue dashboard.
8. Priority and emergency flagging.
9. Digital patient record storage and search functionality.
10. Basic reporting functionality.

Out of Scope

11. Online or cloud-based functionality.
12. Mobile application development.

13. Integration with national health database systems (e.g. NHLS, HPRS).

8. Preliminary Feasibility Assessment

Technical Feasibility

The system can be developed using C# Windows Forms, a local database, and the programming, database, and systems analysis knowledge acquired during the course. All required development tools are available.

Operational Feasibility

The system is designed to be simple enough for reception staff with basic computer literacy to operate confidently from day one. It does not require internet connectivity, making it viable for facilities with unreliable network infrastructure.

Schedule Feasibility

The system is scoped appropriately to be completed within the academic term using proper project planning and teamwork.

9. Initial Requirements

Functional Requirements

- 14. The system must allow patient registration and profile management.
- 15. The system must assign queue numbers automatically upon patient arrival.
- 16. The system must display a live queue dashboard with waiting time tracking.
- 17. The system must allow flagging of emergency and priority patients.
- 18. The system must store, retrieve, and update digital patient records and visit histories.
- 19. The system must generate administrative and operational reports.

Non-Functional Requirements

- 20. The system must be easy to use for staff with basic computer literacy.
- 21. The system must store patient data securely.
- 22. The system must operate without internet connectivity.

10. Initial Risks

Risk	Likelihood	Mitigation
Team member unavailable	Medium	Reallocate tasks among remaining team members
Technical difficulties during development	Medium	Conduct research and seek lecturer guidance

Delays in completing tasks	Medium	Monitor progress and follow the project schedule
Data loss during development	Low	Perform regular backups of all project files

11. Preliminary Feature Backlog

Feature	Priority
User Login	High
Patient Registration	High
Queue Number Assignment	High
Queue Dashboard	High
Patient Record Management	High
Priority / Emergency Flagging	Medium
Visit History Tracking	Medium
Report Generation	Medium
Search Functionality	Medium
Data Backup	Low

12. Expected Benefits

The implementation of Apex is expected to deliver significant improvements across patient care, staff efficiency, and facility management. By replacing paper-based queuing with an automated queue management system, the facility will reduce patient waiting times and provide greater transparency into the queue for both staff and patients.

The digitisation of patient records will eliminate the recurring problem of lost or misplaced files, ensuring that healthcare workers always have access to accurate patient history at the point of care. This is particularly impactful for patients managing chronic conditions such as hypertension, diabetes, or tuberculosis, where medication history and previous diagnoses are critical to delivering safe and effective treatment.

Administrative staff will benefit from reduced manual workloads, faster patient registration, and instant search and retrieval of records. Management will gain access to operational reports that support better resource planning and decision-making. Overall, Apex is expected to improve the quality and consistency of service delivery at the facility while reducing the administrative burden on already stretched healthcare personnel.

13. Team Responsibilities

Team Member	Primary Responsibility
Jongihlati SA	Build and implement the application features.
Mtungwa ST	Plan, coordinate, and keep the project on track.
Mdubo N	Gather requirements and define how the system should work.
Mathebula A	Test the system and report bugs.
Zulu Y	Write and maintain all project documents.
Khwela JM	Design the app's look, layout, and user experience.

Lecturer Approval Section

Project Status:

- ☐ Approved
- ☐ Approved with Changes
- ☐ Rejected

Comments:

Lecturer Signature: _____

Date: _____